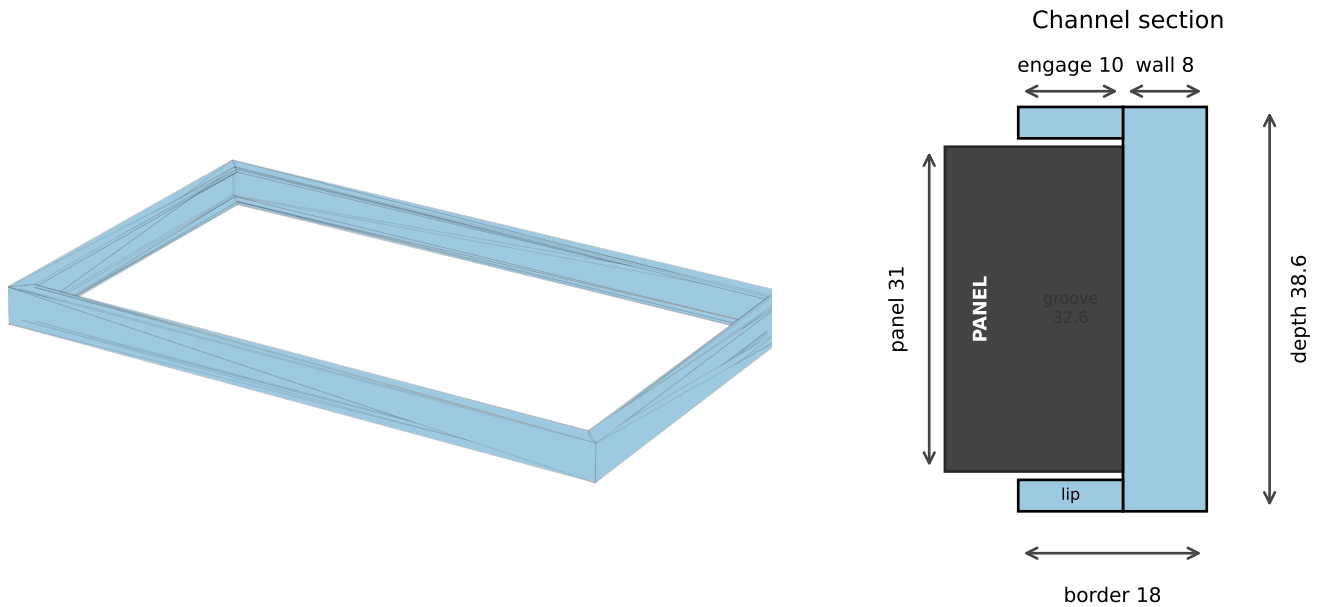


27" DISPLAY TRANSPORT FRAME

Design package — overview



Assembled frame (reference)

KEY SPECIFICATION

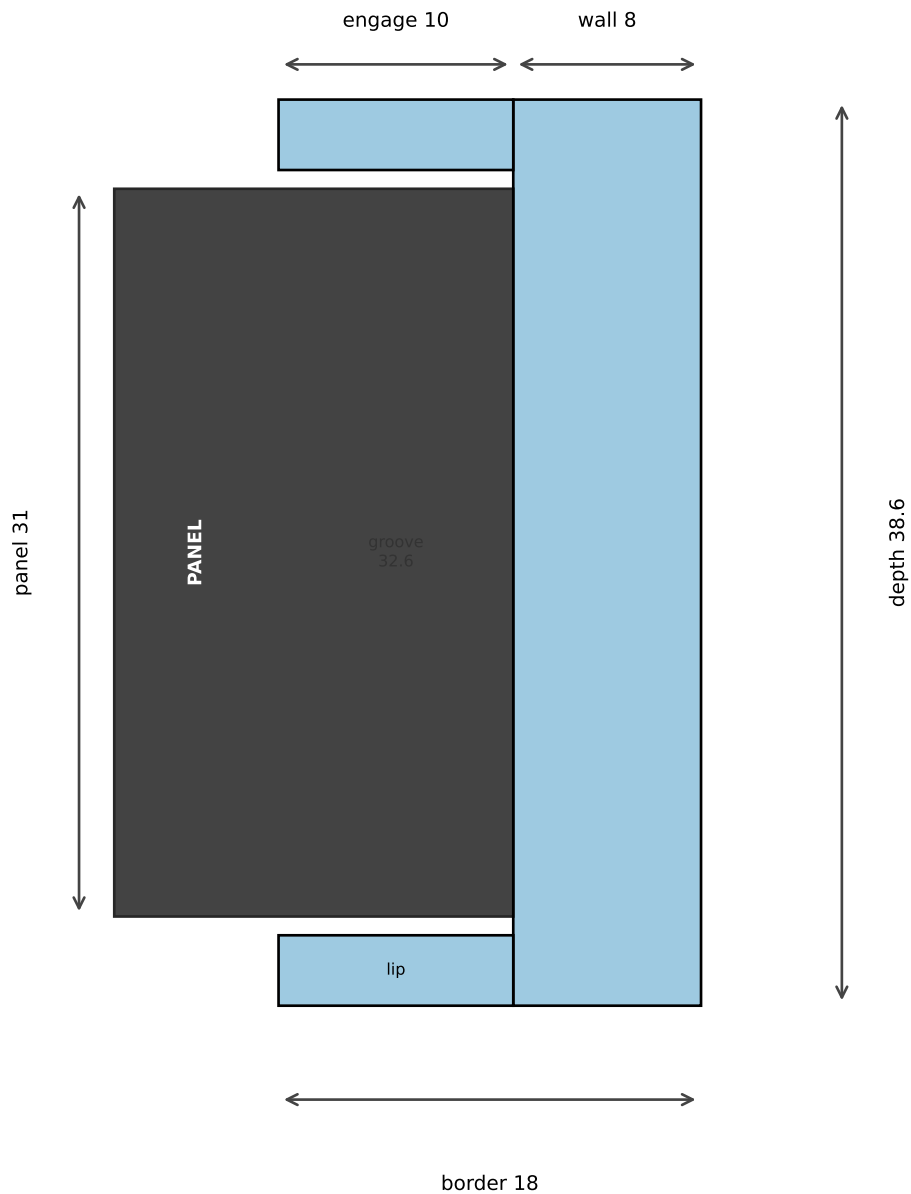
Target display	Apple Studio Display 27" (no stand / VESA)
Panel W x H x D	623 x 362 x 31 mm
Frame outer	640.2 x 379.2 x 38.6 mm
Border width	18 mm (wall 8 + engage 10)
Visible aperture	604.2 x 343.2 mm
Fit clearance	0.6 mm in-plane / 0.8 mm per face
Print bed assumed	220 mm (210 usable)
Pieces	4 corners + 6 top/bottom + 2 side = 12 total
Material (recommended)	PETG / ASA, 4 walls, 30-40% infill
Frame volume	736 cm ³ (~910 g PLA / ~880 g PETG)

C-channel 'picture frame': front + back lips clamp the bezel faces; the outer wall takes edge impacts. The panel edge sits in a groove running the full perimeter. The ring is split into bed-sized pieces joined by Ø3 alignment pins + adhesive.

FIRST: print corner_TR only and test-fit on a real screen corner before printing the full set.

27" DISPLAY TRANSPORT FRAME

Sheet 1 — Channel cross-section



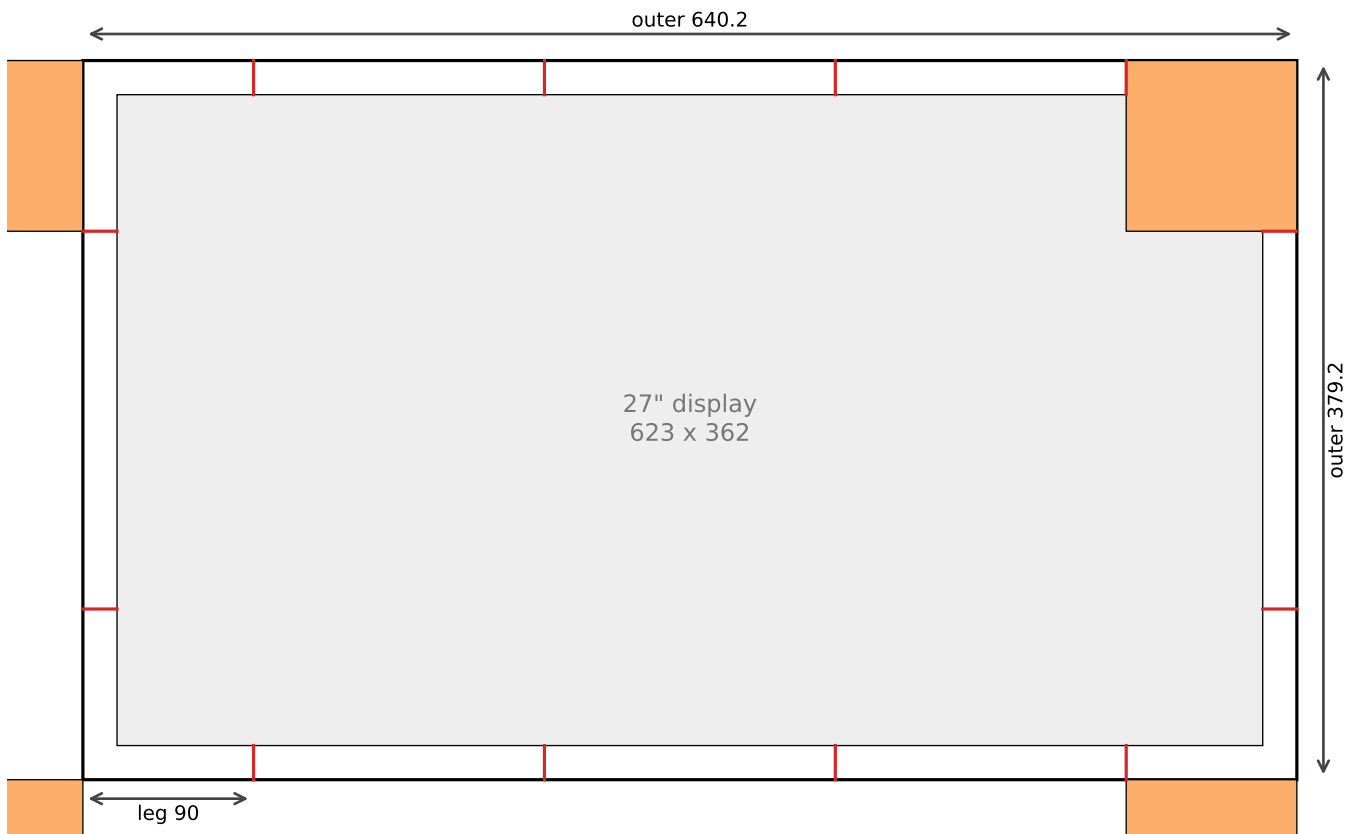
Section is constant around the entire perimeter (mitred at corners).

Tuning:

- `clr / clr_z` – grip tightness (raise if panel won't seat)
- `engage` – how far lips wrap the bezel (also groove depth)
- `wall` – outer impact wall thickness
- `lip_z` – front/back lip thickness

27" DISPLAY TRANSPORT FRAME

Sheet 2 — Frame plan & segmentation

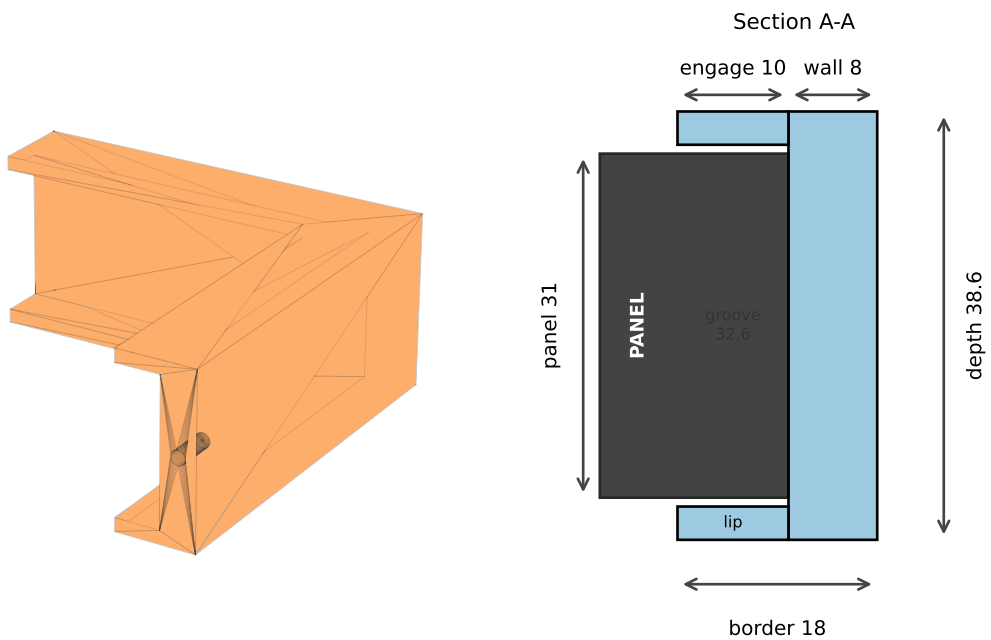
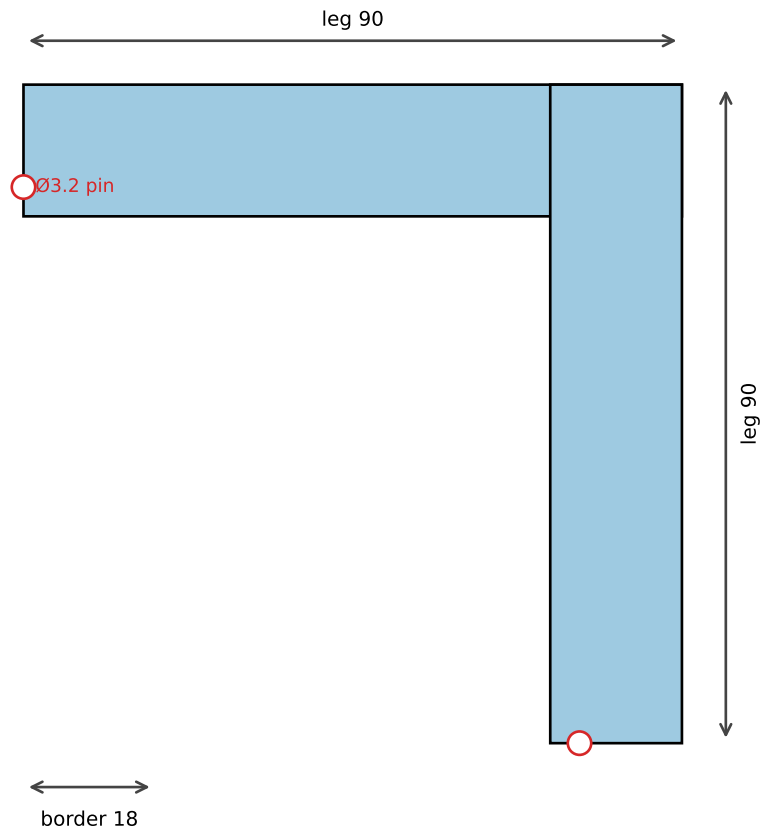


Orange = 4 corner pieces · Red = cut planes · 3 segment(s) per top/bottom run, 1 per side run

27" DISPLAY TRANSPORT FRAME

Sheet 3 — Corner piece (x4: TR, TL, BR, BL)

Front view (looking along screen axis)

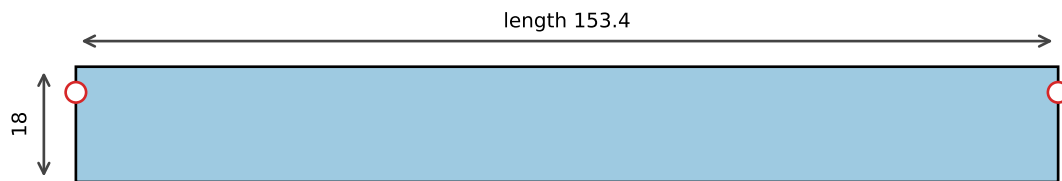


Footprint 90 x 90 x 38.6 mm · 2 pin holes per piece · print flat, no supports.

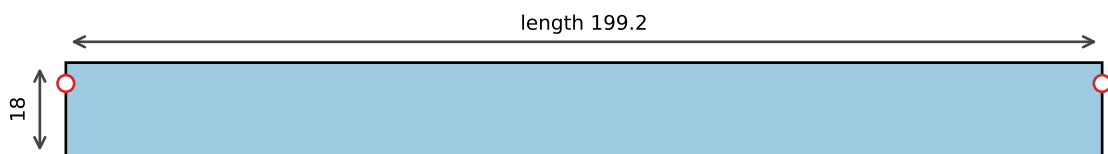
27" DISPLAY TRANSPORT FRAME

Sheet 4 — Edge segments

Top / bottom segment (x6) — pin hole each end



Left / right segment (x2) — shown straightened



All edge segments are border 18 mm x depth 38.6 mm in section (same channel as the corners). Lay flat, depth standing up.

27" DISPLAY TRANSPORT FRAME

Sheet 5 — Bill of materials & cut list

Qty	Part	STL file	Size (mm)
4	Corner (L)	corner_TR/TL/BR/BL.stl	90 x 90 x 38.6
3	Top segment	edge_top_1..3.stl	153 x 18 x 38.6
3	Bottom segment	edge_bot_1..3.stl	153 x 18 x 38.6
1	Right segment	edge_right_1..1.stl	199 x 18 x 38.6
1	Left segment	edge_left_1..1.stl	199 x 18 x 38.6
~12	Alignment pin	pin.stl (or Ø3 rod)	Ø3.0 x 17
1	Whole-ring preview	frame_full_REFERENCE.stl	DO NOT print whole

Total printed body pieces: 12 Joints: 12 Est. filament: ~910 g PLA / ~880 g PETG

CONSUMABLES / HARDWARE

- ~1.0 kg PETG or ASA filament (structural; PLA OK for test fits only)
- Ø3 mm pins: printed pin.stl, or cut 3 mm rod / filament to 17 mm
- Epoxy or CA adhesive for the joints (frame is a permanent ring)
- Closed-cell foam strip to bond the outer wall to the case shell
- Optional: M3/M4 L-brackets to bolt the frame to the case ribs

27" DISPLAY TRANSPORT FRAME

Sheet 6 — Print settings & assembly

PRINT SETTINGS

- Material: PETG / ASA / ABS (impact + heat). PLA for test fits only.
- Orientation: lay flat, 38.6 mm depth standing up -> pin holes horizontal, no supports.
- Perimeters: 4 · Top/bottom layers: 5 · Infill: 30-40% gyroid.
- Layer height: 0.2-0.28 mm · nozzle 0.4-0.6 mm.
- Print one corner first and test-fit before committing the full set.

ASSEMBLY

1. Dry-fit all pieces around the display with pins in the Ø3.2 holes.
2. Check the ring closes square and the lips grip evenly all round.
3. Bond each joint with epoxy/CA; pins keep the faces aligned.
4. Seat the assembled frame in the Rimowa shell.
5. Bond closed-cell foam to the 8 mm outer wall, or bolt L-tabs through it into the case ribs.

MEASURE BEFORE FINAL PRINT

- Exact panel edge thickness (bezels taper) -> set panel_d.
- Rimowa internal depth must clear frame depth 38.6 mm + foam.
- Case internal W x H must clear frame outer 640 x 379 mm.

RETUNE (frame.scad / generate_frame.py)

- Different screen: panel_w / panel_h / panel_d.
- Grip: clr, clr_z. Protection: wall, engage. Printer: bed.
- Re-run: python3 generate_frame.py (STLs) and python3 design_pdf.py (this document).